

## HDOC – 35<sup>th</sup> Day

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### Problem Statement:

Write a menu-driven C program to read a variable-length square integer array and perform the following operations using separate functions:

1. Determine whether all the elements on its main diagonal are distinct.
2. Find the sum of its main diagonal elements.
3. Display its elements using a diagonal traversal. Begin with the element in the top-left corner and traverse successive diagonals alternately in the upward-right and downward-left directions.
4. Exit.

### Algorithm:

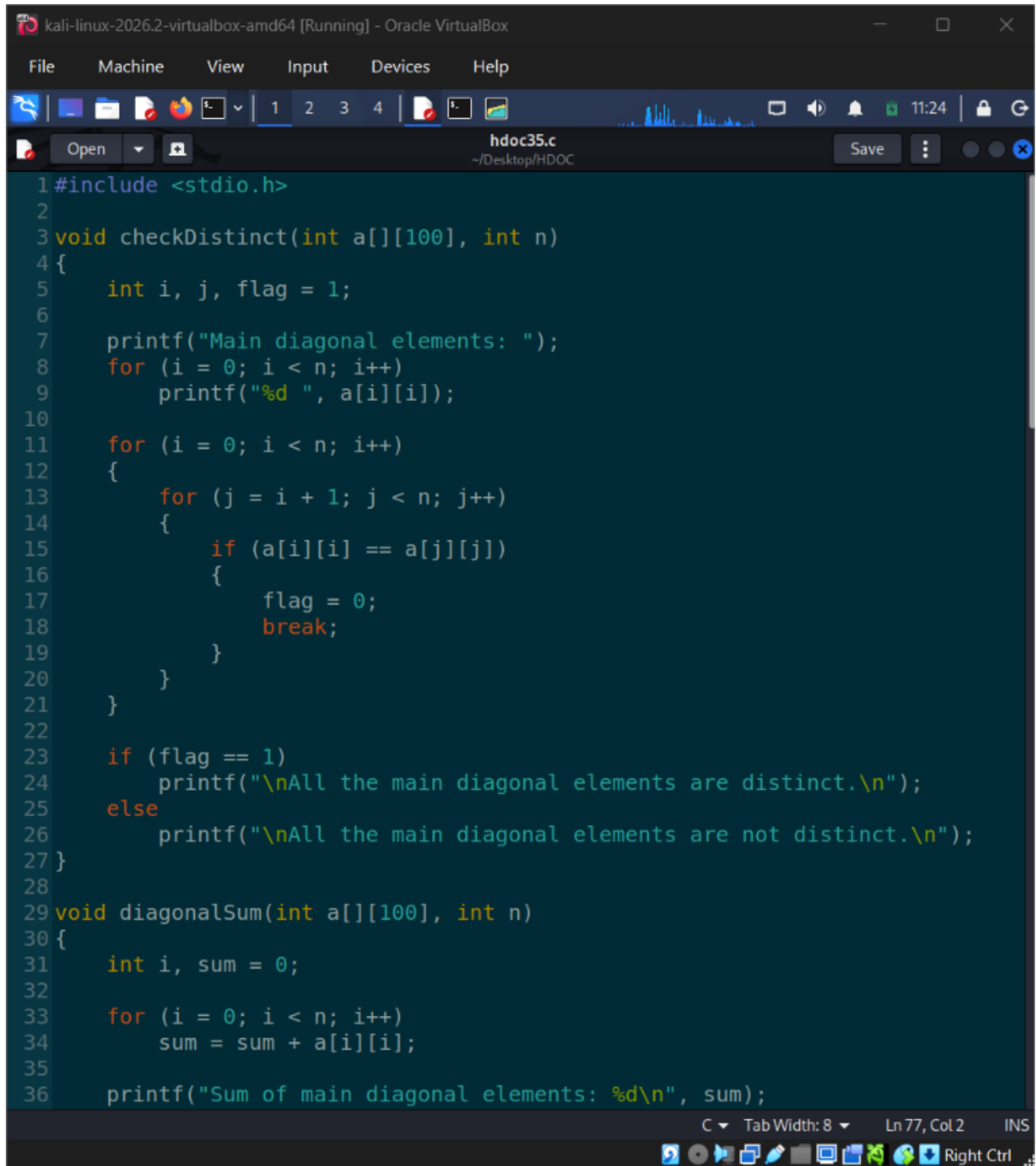
1. Start
2. Input the size  $n$  of the square matrix.
3. Input all elements of the  $n \times n$  matrix.
4. Display the menu:
  - Check whether main diagonal elements are distinct.
  - Find sum of main diagonal elements.
  - Display diagonal traversal.
  - Exit.
5. Input the user's choice.
6. Use a switch statement:
  - Choice 1: Compare the main diagonal elements  $a[i][i]$ . If no two elements are equal, display “All diagonal elements are distinct.”

- Choice 2: Add all main diagonal elements  $a[i][i]$  and display the sum.
  - Choice 3: Traverse the matrix diagonal by diagonal, changing direction alternately, and display the elements.
  - Choice 4: Exit the program.
  - Otherwise, display “Invalid choice.”
7. Repeat the menu until the user selects 4.
8. Stop.

**Test Case Table:**

| Test Case | Input                    | Choice | Expected output                                   | Actual output                                     |
|-----------|--------------------------|--------|---------------------------------------------------|---------------------------------------------------|
| 1         | 1 2 3<br>4 5 6<br>7 8 9  | 1      | Main diagonal: 1 5 9<br>All elements are distinct | Main diagonal: 1 5 9<br>All elements are distinct |
| 2         | 1 2 3<br>4 5 6<br>7 8 9  | 2      | Sum = 15                                          | Sum = 15                                          |
| 3         | 1 2 3<br>4 5 6<br>7 8 9  | 3      | Diagonal transverse:<br>1 2 4 7 5 3 6 8 9         | Diagonal transverse:<br>1 2 4 7 5 3 6 8 9         |
| 4         | 1 2 3<br>4 1 6<br>7 8 9  | 1      | Main diagonal:1 1 9<br>Elements are not distinct  | Main diagonal:1 1 9<br>Elements are not distinct  |
| 5         | Some valid square matrix | 4      | Program ended                                     | Program ended                                     |
| 6         | Some valid square matrix | 5      | Invalid choice                                    | Invalid choice                                    |

## C Program:

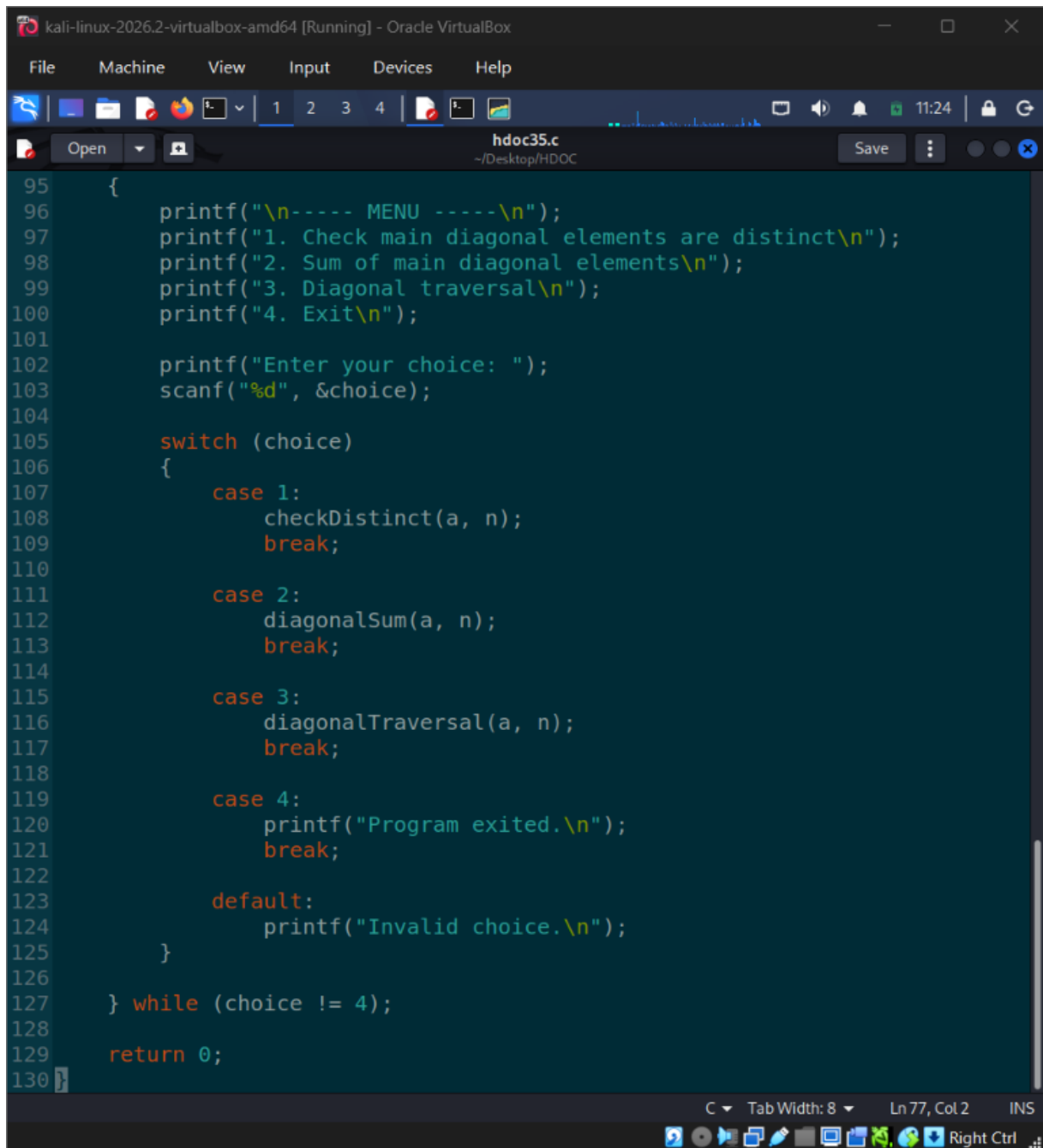
The image shows a terminal window titled 'kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox'. The window has a menu bar with 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help'. Below the menu is a toolbar with icons for file operations and system settings. The main area displays a C program named 'hdoc35.c' located at '~/.Desktop/HDOC'. The code is as follows:

```
1#include <stdio.h>
2
3void checkDistinct(int a[][100], int n)
4{
5    int i, j, flag = 1;
6
7    printf("Main diagonal elements: ");
8    for (i = 0; i < n; i++)
9        printf("%d ", a[i][i]);
10
11    for (i = 0; i < n; i++)
12    {
13        for (j = i + 1; j < n; j++)
14        {
15            if (a[i][i] == a[j][j])
16            {
17                flag = 0;
18                break;
19            }
20        }
21    }
22
23    if (flag == 1)
24        printf("\nAll the main diagonal elements are distinct.\n");
25    else
26        printf("\nAll the main diagonal elements are not distinct.\n");
27}
28
29void diagonalSum(int a[][100], int n)
30{
31    int i, sum = 0;
32
33    for (i = 0; i < n; i++)
34        sum = sum + a[i][i];
35
36    printf("Sum of main diagonal elements: %d\n", sum);
```

The terminal window also shows a status bar at the bottom with 'C', 'Tab Width: 8', 'Ln 77, Col 2', and 'INS'. There is also a 'Right Ctrl' button in the bottom right corner.

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
hdoc35.c
~/Desktop/HDOC Save
35
36     printf("Sum of main diagonal elements: %d\n", sum);
37 }
38
39 void diagonalTraversal(int a[][100], int n)
40 {
41     int d, row, col;
42
43     printf("Diagonal traversal: ");
44
45     for (d = 0; d <= 2 * n - 2; d++)
46     {
47         if (d % 2 == 0)
48         {
49             row = (d < n) ? d : n - 1;
50             col = d - row;
51
52             while (row >= 0 && col < n)
53             {
54                 printf("%d ", a[row][col]);
55                 row--;
56                 col++;
57             }
58         }
59         else
60         {
61             col = (d < n) ? d : n - 1;
62             row = d - col;
63
64             while (col >= 0 && row < n)
65             {
66                 printf("%d ", a[row][col]);
67                 row++;
68                 col--;
69             }
70         }
71     }
72 }
```

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
hdoc35.c
~/Desktop/HDOC Save
69     }
70     }
71 }
72
73 printf("\n");
74 }
75
76 int main()
77 {
78     int a[100][100];
79     int n, i, j, choice;
80
81     printf("Enter the size of square matrix: ");
82     scanf("%d", &n);
83
84     printf("Enter the matrix elements:\n");
85
86     for (i = 0; i < n; i++)
87     {
88         for (j = 0; j < n; j++)
89         {
90             scanf("%d", &a[i][j]);
91         }
92     }
93
94     do
95     {
96         printf("\n----- MENU ----- \n");
97         printf("1. Check main diagonal elements are distinct\n");
98         printf("2. Sum of main diagonal elements\n");
99         printf("3. Diagonal traversal\n");
100        printf("4. Exit\n");
101
102        printf("Enter your choice: ");
103        scanf("%d", &choice);
104
```



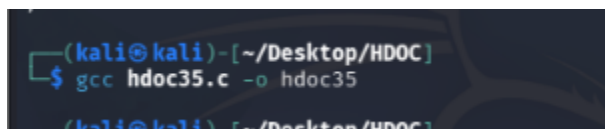
The screenshot shows a Kali Linux virtual machine window titled "kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox". The window has a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". Below the menu bar is a toolbar with icons for file operations and system settings. The main area is a code editor displaying the source code of a C program named "hdoc35.c" located at "~/Desktop/HDOC". The code is as follows:

```
95  {
96      printf("\n----- MENU ----- \n");
97      printf("1. Check main diagonal elements are distinct\n");
98      printf("2. Sum of main diagonal elements\n");
99      printf("3. Diagonal traversal\n");
100     printf("4. Exit\n");
101
102     printf("Enter your choice: ");
103     scanf("%d", &choice);
104
105     switch (choice)
106     {
107         case 1:
108             checkDistinct(a, n);
109             break;
110
111         case 2:
112             diagonalSum(a, n);
113             break;
114
115         case 3:
116             diagonalTraversal(a, n);
117             break;
118
119         case 4:
120             printf("Program exited.\n");
121             break;
122
123         default:
124             printf("Invalid choice.\n");
125     }
126
127     } while (choice != 4);
128
129     return 0;
130 }
```

The code editor has a status bar at the bottom showing "C", "Tab Width: 8", "Ln 77, Col 2", and "INS". The system tray at the bottom right shows various icons and the text "Right Ctrl".

Compile the program using:

```
gcc hdoc35.c -o hdoc35
```



The screenshot shows a terminal window with the following text:

```
(kali@kali)-[~/Desktop/HDOC]
$ gcc hdoc35.c -o hdoc35
(kali@kali)-[~/Desktop/HDOC]
```

Compile the program using:

`./hdoc35`

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle Virtual
File Machine View Input Devices H
1 2 3 4
kali@kali: ~
Session Actions Edit View Help
(kali@kali)~[~/Desktop/HDOC]
$ ./hdoc35
Enter the size of square matrix: 3
Enter the matrix elements:
1 2 3 4 5 6 7 8 9

—— MENU ——
1. Check main diagonal elements are distinct
2. Sum of main diagonal elements
3. Diagonal traversal
4. Exit
Enter your choice: 1
Main diagonal elements: 1 5 9
All the main diagonal elements are distinct.

—— MENU ——
1. Check main diagonal elements are distinct
2. Sum of main diagonal elements
3. Diagonal traversal
4. Exit
Enter your choice: 4
Program exited.
```

```
(kali@kali)~[~/Desktop/HDOC]
$ ./hdoc35
Enter the size of square matrix: 3
Enter the matrix elements:
1 2 3 4 5 6 7 8 9

—— MENU ——
1. Check main diagonal elements are distinct
2. Sum of main diagonal elements
3. Diagonal traversal
4. Exit
Enter your choice: 2
Sum of main diagonal elements: 15

—— MENU ——
1. Check main diagonal elements are distinct
2. Sum of main diagonal elements
3. Diagonal traversal
4. Exit
Enter your choice: 4
Program exited.
```

```
kali
Session Actions Edit View Help
(kali@kali)~[~/Desktop/HDOC]
$ ./hdoc35
Enter the size of square matrix: 3
Enter the matrix elements:
1 2 3 4 1 6 7 8 9

—— MENU ——
1. Check main diagonal elements are distinct
2. Sum of main diagonal elements
3. Diagonal traversal
4. Exit
Enter your choice: 1
Main diagonal elements: 1 1 9
All the main diagonal elements are not distinct.

—— MENU ——
1. Check main diagonal elements are distinct
2. Sum of main diagonal elements
3. Diagonal traversal
4. Exit
Enter your choice: 4
Program exited.
```

```
(kali@kali)~[~/Desktop/HDOC]
$ ./hdoc35
Enter the size of square matrix: 3
Enter the matrix elements:
1 2 3 4 5 6 7 8 9

—— MENU ——
1. Check main diagonal elements are distinct
2. Sum of main diagonal elements
3. Diagonal traversal
4. Exit
Enter your choice: 3
Diagonal traversal: 1 2 4 7 5 3 6 8 9

—— MENU ——
1. Check main diagonal elements are distinct
2. Sum of main diagonal elements
3. Diagonal traversal
4. Exit
Enter your choice: 4
Program exited.
```

```
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc35
Enter the size of square matrix: 3
Enter the matrix elements:
1 2 3 4 5 6 7 8 9

—— MENU ——
1. Check main diagonal elements are distinct
2. Sum of main diagonal elements
3. Diagonal traversal
4. Exit
Enter your choice: 4
Program exited.
```

```
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc35
Enter the size of square matrix: 3
Enter the matrix elements:
1 2 3 4 5 6 7 8 9
9
9

—— MENU ——
1. Check main diagonal elements are distinct
2. Sum of main diagonal elements
3. Diagonal traversal
4. Exit
Enter your choice: 5
Invalid choice.

—— MENU ——
1. Check main diagonal elements are distinct
2. Sum of main diagonal elements
3. Diagonal traversal
4. Exit
Enter your choice: 4
Program exited.

(kali@kali)-[~/Desktop/HDOC]
$
```